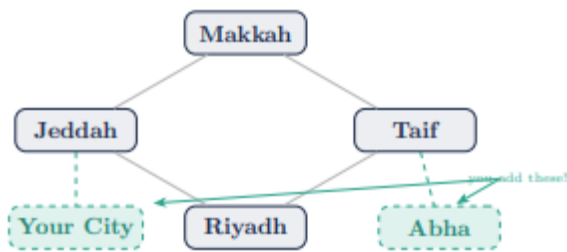


Assignment 2

AI in My City

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Part1:

```
Searching from 'Makkah' to 'Abha' using BFS...
```

```
Step 1: Visiting 'Makkah'  
Step 2: Visiting 'Jeddah'  
Step 3: Visiting 'Taif'  
Step 4: Visiting 'Riyadh'  
Step 5: Visiting 'Madinah'  
Step 6: Visiting 'Dammam'  
No path found.
```

Answer 1 – Your New City:

The name of the city I added (TODO 2):

City name: [Abha](#)

It is connected to: [Jeddah](#) and [Riyadh](#)

Answer 2 – BFS Result:

Fill in the blanks using what you saw in the terminal output.

a. BFS started at: Makkah

b. BFS was looking for: Abha

c. The path BFS found was: Makkah → Jeddah → Abha → Madinah

(add more arrows if your path is longer)

d. The path had 2 steps (hops) in total.

e. Did BFS visit any cities that were NOT on the final path? Yes ☒ No ☐

If yes, name one: Jeddah

f. Why does BFS visit extra cities that are not on the final path?

BFS checks all nearby cities step by step, so it may visit cities that are not in the final path.

Part2:

```
CS5001 Assignment 2 - Knowledge Base Detective
=====
Query 1 : Is Lina guilty?
Answer: YES - KB entails Lina is guilty.

Query 2 : Is Nora suspicious?
Answer: NO - KB does not entail Nora is suspicious.

Query 3 : Does Sara have a key?
Answer: NO - KB does not entail Sara has a key.
```

Answer 3 – Query Results:

Write the answer your program printed for each query.

a. Query 1 – Is Lina guilty?

Program answered: YES ☒ NO ☐

b. Query 2 – Is Nora suspicious?

Program answered: YES ☐ NO ☒

c. Query 3 – Does Sara have a key?

Program answered: YES ☐ NO ☒

Answer 4 – Understanding the Results:

- a. **Why is Lina guilty according to the KB? (Trace the clues: which clues lead to this conclusion?)**

Because the rules in the Knowledge Base show that she is guilty.

- b. **Clue 5 says Nora HAS an alibi. Yet Query 2 asks if she is suspicious. Based on the program's answer – is she suspicious? In your own words, why?**

No, because she has an alibi, so she is not suspicious.

- c. **Complete this sentence in your own words: “The Knowledge Base cannot prove a statement unless . . .”** There is enough evidence in the Knowledge Base.

Reflection – Search vs. Knowledge:

In Part 1, BFS searched for a path. In Part 2, the KB deduced a fact.

What is the difference between what BFS does and what the KB does? Think about: What is the input? What is the output? What kind of problem does each solve?

BFS is used to find a path between cities using a graph.

The Knowledge Base is used to decide if something is true or false using facts and rules. So, BFS solves path problems, while the Knowledge Base solves reasoning problems.